PRESENTED BY

GENERATIVE GAINS

Thermal Trace

QATAR FIRST, GCC NEXT

Know what a failing fridge is costing before the gate does.

Cold-chain intelligence: physical risk and commercial value, in one reserve.



• CITED · FAO / UNEP 2024

A third of our food never gets eaten.

12.3 SDG 12.3 · halve food waste, cut supply-chain loss

2 SDG 2 · zero hunger

13 SDG 13 · climate action

13%

of food is **lost** between harvest and retail.

19%

more is **wasted** after it reaches retail and homes.

8–10%

of global greenhouse-gas emissions come from food loss and waste.

CITED · TASMU · QATAR NFSS 2030

An import-dependent region, a brutal summer.

Climate and imports



45 °C

Imports by sea and land, customs holds and border queues, across a summer that passes 45 °C. A fridge that fails at noon can spoil a load in hours.

Policy tailwind

TASMU analytics and Qatar's Food Security Strategy target efficient supply chains and less food waste. Thermal Trace works on the loss in transit.

NATIONAL FOOD SECURITY STRATEGY 2030

Food waste		−50%
Food loss		−30%

THERMAL DEBT — Maritime import + Customs hold + Border queue + 45 °C summer = Lost shelf life

Thermal Trace is the **operator-level layer** that complements national systems.

○ COMMERCIAL RESERVE · DESIGNED, STAGED IN THE PILOT

Safe is not the same as sellable.

Thermal Reserve THE PHYSICAL CLOCK	Commercial Reserve THE COMMERCIAL CLOCK
FOCUS Product health: “Can it still be consumed?”	FOCUS Economic life: “Can it still earn its value?”
CLOCK Production → Transport → Deterioration → Shelf-life boundary	CLOCK Arrival → Customer exposure → Sell-through → Replenishment
GOAL Protect physical life. ● CALCULATED	GOAL Protect economic life. ● PREDICTED

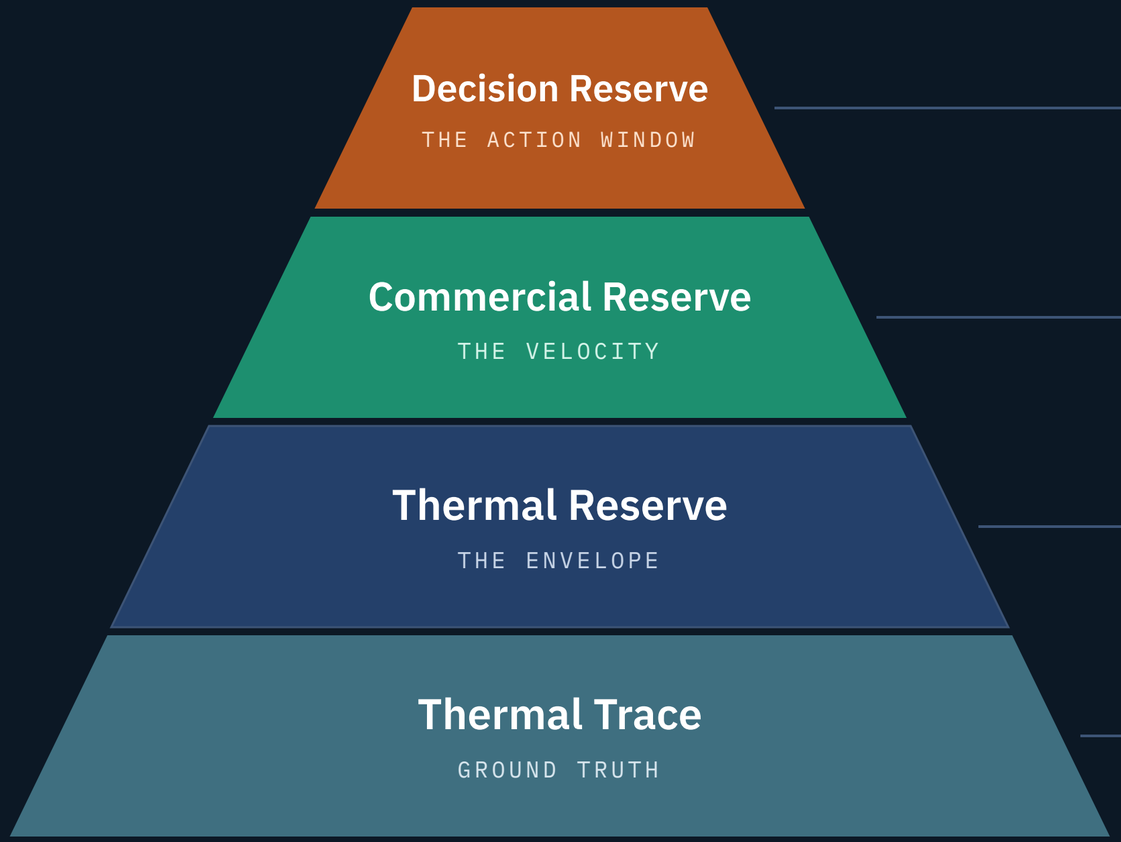
Health creates the possibility of selling. Demand creates the possibility of realising value.
Commercial Reserve lives where the two meet.

$$CR = Q_r \times P_r \times V_r$$

Q_r quantity realistically realisable
P_r probability of sale
V_r expected value per unit

LAYER 3 DESIGNED · STAGED IN THE PILOT

The anatomy of value: four reserves, one stack.



LAYER 4 · DECISION RESERVE · OPTIMIZED

The time left to act before the good options disappear.

LAYER 3 · COMMERCIAL RESERVE · DESIGNED

The economic runway: value that can still be realised.

$$CR = Q_r \times P_r \times V_r$$

LAYER 2 · THERMAL RESERVE · CALCULATED

The physical budget left before quality is compromised.

LAYER 1 · THERMAL TRACE · MEASURED

A continuous, timestamped memory of temperature exposure and stress.

Six signals. Three numbers that matter.

● TEMPERATURE

DHT11 today · DS18B20 next

The demo node reads one point. Ceiling and floor probes are the next step, so a warm layer shows up instead of hiding in an average.

● HUMIDITY

DHT11 today · SHT3x next

Rising humidity shows whether an open door is wetting the cartons.

○ SHOCK & TILT · SIMULATED

3-axis MEMS · designed

Bruising from a dropped pallet is damage a thermometer never sees.

○ GASES · DESIGNED

Metal-oxide · VOC, C_2H_4

Ethylene is the fruit itself reporting that it has started to ripen.

○ SUB-SURFACE · DESIGNED

NIR · 700–2500 nm

Finds internal bruising before anything is visible from outside.

○ SURFACE · DESIGNED

RGB camera · edge vision

Colour shift, mould, and a count of what is actually on the shelf.

THEY FEED → Deterioration detection · Remaining Useful Time · Thermal exposure E_T ● CALCULATED

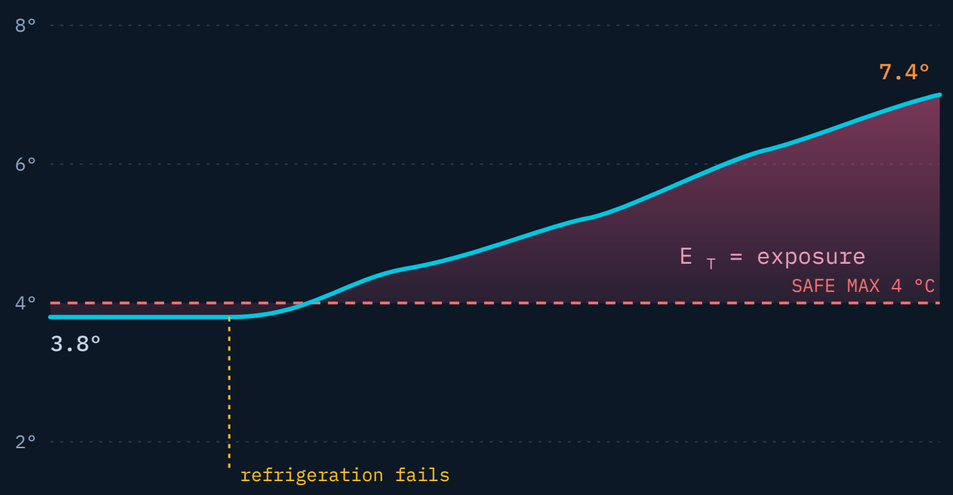
Temperature and humidity flow from a real sensor node today (DHT11, about ±2 °C); shock is simulated and the rest are the designed next layer. All publish the same message, so adding one changes nothing downstream.

○ EXAMPLE SCENARIO

The label says 7 days. The trace says 4.5.

LAYER 1 · THERMAL TRACE · CARGO TEMPERATURE

● MEASURED



THERMAL EXPOSURE · °C·MIN

$$E_T = \Sigma \max(0, T_i - T_{safe}) \cdot \Delta t$$

DETERIORATION RATE · ARRHENIUS

$$k(T) = k_{ref} \cdot e^{(E_a/R)(1/T_{ideal} - 1/T)}$$

LAYER 2 · THERMAL RESERVE · SAFE LIFE LEFT

$$D = \Sigma\, k(T_i) \cdot \Delta t_i \rightarrow \text{reserve} = L \cdot (1 - D)$$

LABEL DATE		7.0 d
THERMAL RESERVE		4.5 d

● CALCULATED Deterministic Python. No AI model touches these numbers.

AI ON A LEASH
NEITHER MAY INVENT
A NUMBER

Laya SYSTEM 1 · LOCAL · OFFLINE ● PREDICTED

Answers typed questions in one pass. Never generates text; corroborates the decision, never overrides it. Base model, confidence shown as uncalibrated.

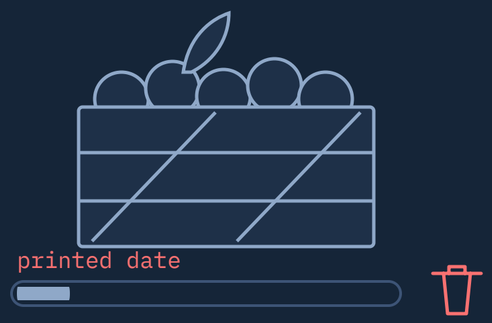
DeepSeek V4.1 Flash SYSTEM 2 · CLOUD ● AI-EXPLAINED

Explains the calculated facts in plain words. May rank only feasible options; never invents a temperature, ETA or cost.

Redefining the perishable asset.

The biological clock

DEFENSIVE · TODAY

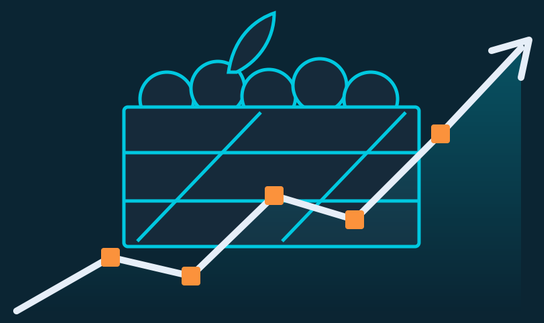


Legacy model: inventory is a degrading liability, managed by a static printed expiry date.

Result: late markdowns and write-offs.

The commercial trace

OFFENSIVE · THERMAL TRACE



Thermal Trace model: inventory is an asset with a live value, priced from its real thermal history and live demand.

Result: value recovered before it is lost.

○ COUNTDOWN · EXAMPLE · COMMERCIAL RESERVE DESIGNED

Two clocks, one platform.

Thermal Trace tells us what is happening to the product. Two reserves read it.

PHYSICAL CLOCK

● CALCULATED

Thermal Reserve

How much physical margin remains.

45 min left

For 3PLs, fleets and warehouses. Sold as a platform fine-tuned to each customer: sensors, calibration and integration.

3PLs | Fleets | Warehouses

COMMERCIAL CLOCK

● PREDICTED

Commercial Reserve

How much economic opportunity remains, and what we should do about it.

POS · ERP · WMS → route → price to clear

For hypermarkets and HORECA. Sold as SaaS: joins the physical state with POS, ERP and WMS to route stock, and price it to clear.

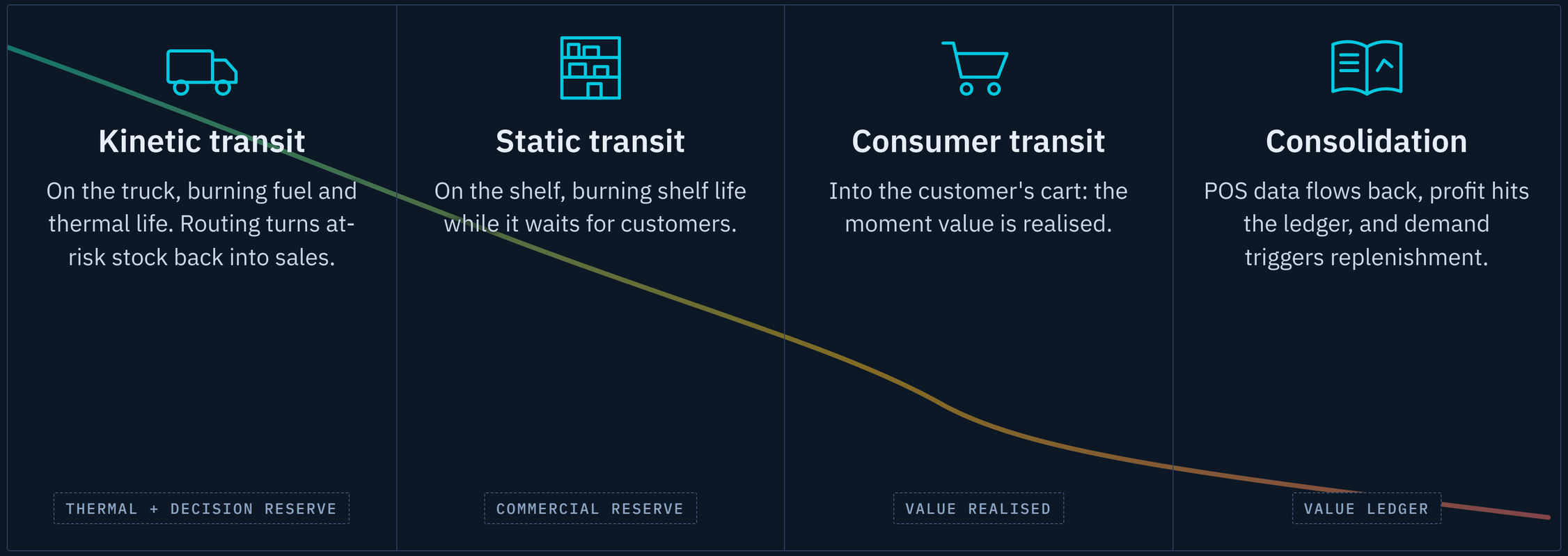
Hypermarkets | HORECA

ENTRY POINT · THERMAL RESERVE · TUNED PLATFORM ——→ CONTROLLED EXTENSION · COMMERCIAL RESERVE · SAAS

ILLUSTRATIVE VALUE CURVE · DESIGNED

In transit until the profit is booked.

The trace follows the value from farm to ledger, not just from dock to dock.



Kinetic transit

On the truck, burning fuel and thermal life. Routing turns at-risk stock back into sales.

THERMAL + DECISION RESERVE

Static transit

On the shelf, burning shelf life while it waits for customers.

COMMERCIAL RESERVE

Consumer transit

Into the customer's cart: the moment value is realised.

VALUE REALISED

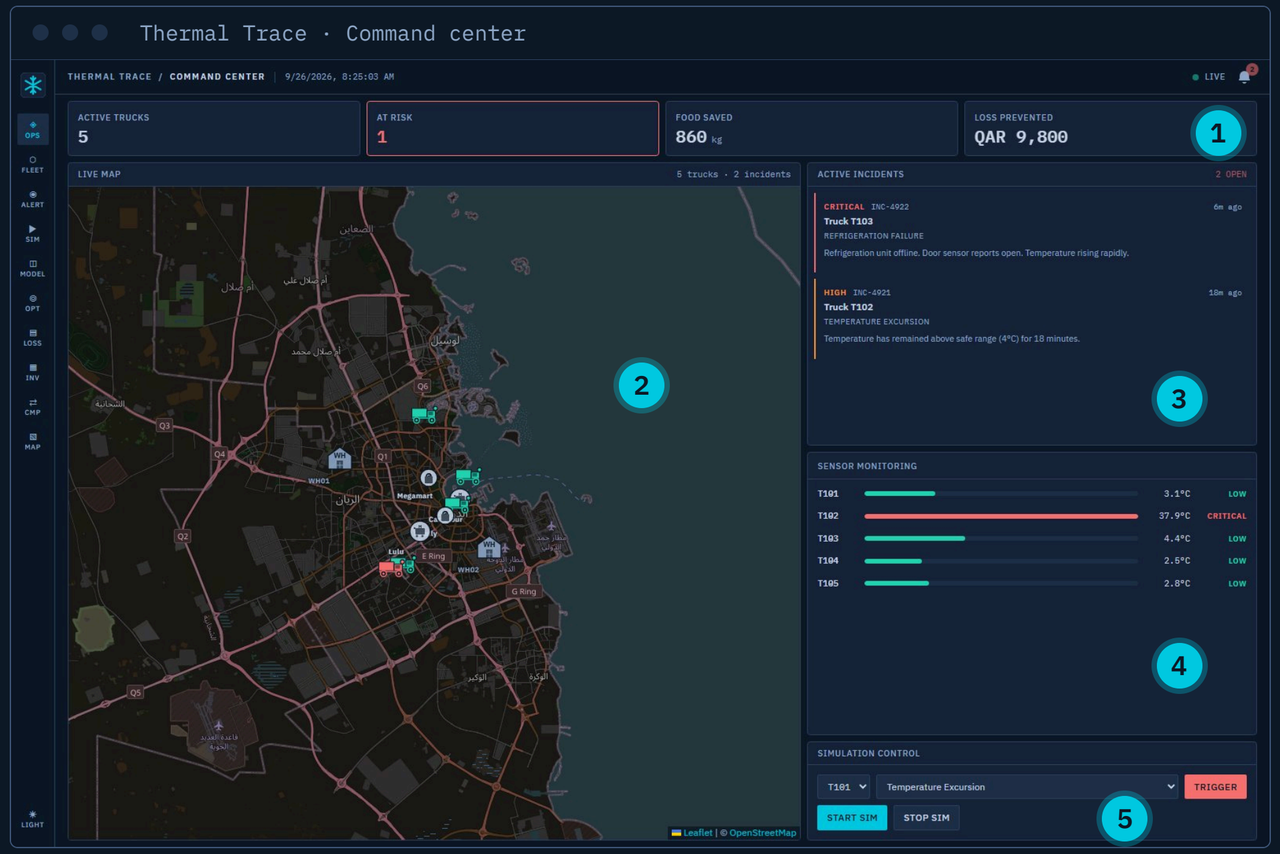
Consolidation

POS data flows back, profit hits the ledger, and demand triggers replenishment.

VALUE LEDGER

○ LIVE DEMO · SIMULATED DATA

One screen for the whole cold chain.



1. **Headline figures**
   Active trucks, at risk, food saved, loss prevented.
2. **Live map**
   Every truck, warehouse and store, coloured by risk.
3. **Active incidents**
   Ranked by severity, with the cause in plain words.
4. **Sensor monitoring**
   Each truck's cargo temperature against its safe range.
5. **Simulation control**
   Trigger a failure and watch the system respond.

Data → Condition → Prediction → Optimization → Decision → Food loss

○ EXAMPLE SCENARIO

The window to act, and the move to make.

$$\min(C_{transport} + C_{food\text{-}loss} + C_{delay})$$

SUBJECT TO

- ETA ≤ remaining thermal reserve
- Quantity ≤ available capacity
- Storage temperature compatible

AI — Corroborates — Explains — Never invents a number

● REEFER Compressor failing — **Thermal reserve 45 min**

DESTINATION	ETA	FITS 45 MIN
Hypermarket Planned route	90 min	✗ spoils
Nearby buyer Re-routed	20 min	✓ fits

● OPTIMIZED **REROUTE**

Decision reserve ≈ 25 min to commit. The optimizer picks; the AI only explains why.

○ ILLUSTRATIVE · PRODUCTS NAMED FOR REFERENCE

Banana fails cold. Ice cream fails warm.

SIGNAL	🍌 13–14 °C · 90–95 % RH Too cold browns it, ethylene ripens it	🍦 –18 °C or colder Any thaw-refreeze ruins texture	Gate check only One logger, read after delivery	Bought trackers In transit vs handheld at QA
● TEMPERATURE DHT11 today · DS18B20 next	Below 13 °C = chilling injury: peel turns grey days later, after it is sold on.	A rise above –18 °C grows ice crystals; it refreezes gritty and nobody can tell from outside.	Excursion found at the gate or by the customer. Load already damaged.	Tive Solo 5G, Sensitech TempTale GEO X, ELPRO LIBERO
● HUMIDITY DHT11 today · SHT3x next	Too dry and skins shrivel and scar; too wet and cartons go soft.	A door left open lets moist air in: frost on cartons proves a warm-up.	Soggy or dry cartons noticed only when unloading.	Tive Solo 5G, ELPRO LIBERO Gx
○ SHOCK & TILT · SIMULATED 3-axis MEMS · designed	A dropped pallet bruises; bruises appear 1–2 days later.	Crushed tubs and popped lids; tilt lets softened product leak.	Damage claims argued with no record of where or when.	Tive Solo 5G (12 g shock), Controlant Saga (vibration)
○ GASES · DESIGNED Metal-oxide · VOC, C_2H_4	Ethylene rising: the load started ripening in transit, so sell it first.	*Not used. Sealed product, and MOx sensors are not rated this cold.*	Over-ripe load discovered at the store, too late to reroute.	*Handheld only: Felix F-900 ethylene analyser*
○ SUB-SURFACE · DESIGNED NIR · 700–2500 nm	Starch turning to sugar inside, before the peel shows it.	*Not used. Packaging blocks it; temperature history tells the story.*	Ripeness judged by eye at the dock.	*Handheld only: Felix F-750, VIAVI MicroNIR*
○ SURFACE · DESIGNED RGB camera · edge vision	Peel colour stage, brown spots, crown mould, count on shelf.	Frosted or misshapen packs, open lids, empty facings.	Staff walk the aisle a few times a day.	*No in-transit product; fixed shelf cameras only*

THEY FEED → Deterioration detection · Remaining Useful Time · Thermal exposure E_T ● CALCULATED

From managing biology to managing liquidity.

DIMENSION	LEGACY SUPPLY CHAIN	THERMAL TRACE
THE GOAL	Delay spoilage (defensive)	Maximise yield (offensive)
ASSET VIEW	A degrading physical object	An asset with a live, priced value
INVENTORY RULE	FEFO: first expired, first out, by printed date	LCRFO (planned): lowest Commercial Reserve, first out, by live physics
INTERVENTION	Passive temperature alarms	Optimized routing today; early markdowns planned
WASTE	An invisible operating tax	A quantified, bounded value at risk

HYPOTHESES TO VALIDATE · NOT MARKET PRICES

Tailored to each customer. Sold as SaaS.

THERMAL TRACE PLATFORM · SHARED IP

- Trace engine
- Food-state model
- Data ingestion
- Deterministic governance
- Decision workflows
- Value ledger

PHYSICAL INTELLIGENCE

Thermal Reserve

Configured to the customer's equipment, sensors and products. The deployment is customised; the platform stays reusable.

MARKET INTELLIGENCE

Commercial Reserve

Precise calculations over the retailer's own POS, inventory and ERP data. The repeatable SaaS product.

Pay 1×, recover ≥ 3× every QAR 1 charged saves QAR 3+ of food.

1 · ONE-TIME SETUP

Implementation — QAR 90K

Fixed setup cost. Hardware they need is billed separately, at cost.

→

2 · YEARLY SUBSCRIPTION · PICK ONE

Standard — QAR 120K / yr

One product · live monitoring, alerts and reserves · monthly value report · email support

2 · YEARLY SUBSCRIPTION · PICK ONE

Enterprise — QAR 216K / yr

Both products · customised dashboard · unlimited integrations · SSO and uptime SLA · full support

Nine months to prove the market.

QAR 650,000

≈ US$179,000 · 9 months · market validation

Enough to prove one thing: that operators will **pay to stop losing food in transit.**

12.3 Every kilogram saved is logged as evidence.

WHAT IT PROVES IN 9 MONTHS

01
One pilot site
Thermal Reserve live at a reefer fleet or cold store

02
Avoided loss
Measured on real loads, against a baseline

03
Willingness to pay
A paid pilot or signed letters of intent

04
Expert review
Shelf-life models reviewed by food-safety experts

05
The next round
The evidence to raise it

○ VISION · NOT BUILT

One primitive, many industries.

Any asset that deteriorates, cures or loses viability under the stress of time needs this architecture.

Biopharma ZERO-TRUST COMPLIANCE

PHYSICS	stability budgets
CALCULATES	active-cooling intercepts on the tarmac

Energy THE GRID TRACE

PHYSICS	grid signals, inverter telemetry
CALCULATES	grid flexibility reserve

TRACE → STATE → RESERVE → DECISION

BUILT FIRST FOR THE FOOD COLD CHAIN

Space cargo THE ORBITAL TRACE

PHYSICS	propellant bleed, thermal cycling
CALCULATES	mission reserve for manoeuvres

Defence THE KINETIC TRACE

PHYSICS	munition shock and degradation
CALCULATES	operational readiness reserve

THERMAL TRACE · BY GENERATIVE GAINS

Thank you.

GENERATIVE GAINS

Generative Gains is proving Thermal Trace in the field: that what a failing fridge costs can be measured, and recovered.

01

Najeeb Abdi

MATH · FRONTEND

02

Islambek

HARDWARE ENGINEER

03

Param Anand Trimbake

FRONTEND

04

Ahad Hussain

AI ENGINEER

05

Robin Thomas

BUSINESS · PRODUCT

Generative Gains · Qatar first, GCC next

github.com/Isl-d/Reboot-The-Earth-Cold-Chain-Logistics